Design of Integrated Passive Component for a 1MHz 1kW Half-bridge LLC Resonant Converter

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***Abstract*—The LLC resonant converter has some special characteristics to make it an excellent candidate for front-end DC/DC conversion in a Distributed Power System. Even with advanced topology, to achieve high power density, the passive component design is still the key issue. Passive integration is used to integrate all the passive components within the resonant tank of the LLC resonant converter into one single component – an integrated L-L-C-T. The design approach and loss estimation of the integrated L-L-C-T component for a 1MHz 1kW LLC resonant converter is given in this paper. A prototype is constructed and tested under small signal and in-circuit working conditions.**

Keywords-Integrated L-L-C-T component; LLC Resonant Converter; Passive Integtation;

I. INTRODUCTION

The LLC resonant converter has some special characteristics to make it an excellent candidate for front-end DC/DC conversion in a Distributed Power System. Firstly, an LLC resonant converter can be optimized for high input voltage, which is the normal operating condition. In comparison, for some PWM topologies, such as an asymmetric half bridge or phase shift full bridge converter, a wide duty cycle range is necessary for a wide input voltage range. The efficiency of the converters under normal operating conditions suffers because of the hold up time requirement [1]. Secondly, Zero Voltage Switching (ZVS) turn on and low turn off current can be achieved in a LLC resonant converter, thus the switching loss can be reduced and the switching frequency of the LLC resonant converter can be pushed higher. Thirdly, the secondary filter inductor can be eliminated, so the voltage stress on the secondary rectifier is limited to two times the output voltage, better rectifier diodes can be used and secondary conduction loss can be reduced [2].

The schematic of a half-bridge LLC resonant converter is shown in Fig. 1. Within the dashed lines is the resonant tank of the LLC resonant converter, which consists of the resonant inductor Lr, the resonant capacitor Cr, the magnetizing inductor Lm and the transformer T. To achieve higher power density and reduce the number of magnetic components, an integrated magnetic design is proposed in [3] to integrate the magnetic components of the converter (Lr, Lm and T) into one magnetic core. By sharing the magnetic path, flux ripple can be reduced, thus smaller magnetic core loss can also be achieved.

To further reduce the number of passive components, achieve higher power density, better performance and higher reliability, passive integration is proposed to integrate all passive components within the resonant tank (including the resonant capacitor Cr) into one single component – an integrated L-L-C-T. The design of the integrated L-L-C-T component for a 1MHz 1kW half-bridge LLC resonant converter is given in this paper. The losses of the integrated L-L-C-T component, including magnetic core loss, dielectric loss and winding conduction loss, are estimated. A prototype of the integrated L-L-C-T component is constructed. Small signal test results show that the measured parameter values of the integrated component agree well with the design values. In circuit test results of the integrated L-L-C-T component are also given and compared with the test of discrete passive components.

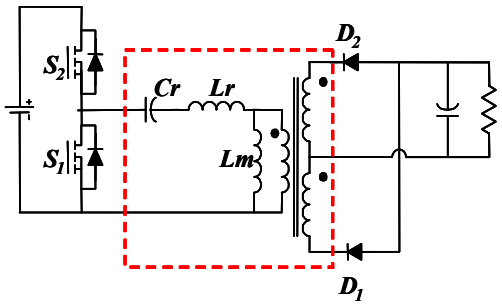

Fig. 1 Schematic of half-bridge LLC resonant converter

II. INTEGRATED L-L-C-T COMPONENT

A generic exploded view of an integrated L-L-C-T component is shown in Fig. 2. It consists of 4 parts: the magnetic core; the leakage layer; the primary winding, whose structure is shown in Fig. 3, it is a sandwich structure with dielectric layer in the middle and two copper layers on both sides; the secondary winding, which is made of copper foil [5-9].

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The magnetic core, primary and secondary winding together can function as a transformer T. The core is air gapped to get the desired magnetizing inductance Lm. The leakage inductance of the transformer can be used as the resonant inductor Lr of the LLC resonant circuit. Usually the inherent leakage inductance is not large enough, so a leakage layer is added between the primary and secondary windings. The leakage inductance value can be changed by adjusting the thickness of the leakage layer. The value of the resonant capacitor Cr depends on the copper trace area of the primary winding and the relative permittivity and thickness of the dielectric material. Thus one single integrated L-L-C-T component can replace all the discrete passive components in the resonant tank of the LLC resonant converter.

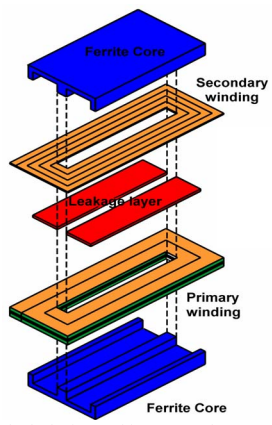

Fig. 2 Exploded view of integrated L-L-C-T component

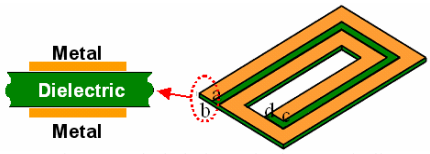

Fig. 3 Exploded view of primary winding

III. Design of LLC Resonant Converter

To obtain the design values and operating conditions for the integrated L-L-C-T component, the design of the LLC resonant converter is considered first. The LLC resonant converter has two resonant frequencies. Lm+Lr and Cr determine the lower resonant frequency; Lr and Cr determine the higher resonant frequency. The higher resonant frequency is in the ZVS region. The converter should be designed to operate around this region [2].

Fig. 4 shows the DC characteristic and three operating regions of the LLC resonant converter. Region 1 and region 2 are ZVS (Zero Voltage Switch) regions, which are preferred for high frequency operation; region 3 is the ZCS (Zero Current Switching) region and the converter should not enter this region. Region 1 and region 2 each has 3 different operating modes for different load conditions. The details about different operating modes of LLC resonant converter and the design of power stage parameters will not be discussed here. They can be found in [2].

After completing the design of the LLC resonant converter, the design values of the integrated L-L-C-T component can be determined. The operating conditions of the integrated L-L-C-T component can be determined by circuit analysis or simulation. Table I shows some of the design parameters of the 1MHz 1kW LLC resonant converter and the design specifications of the integrated L-L-C-T component.

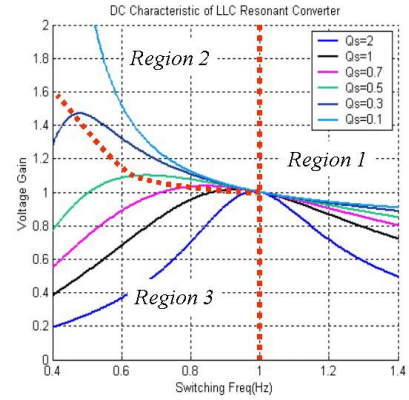

Fig. 4 DC characteristic and operating regions of LLC resonant converter

TABLE I. Operating conditions of LLC resonant converter and design specifications of integrated L-L-C-T

Parameters	Symbol	Value
Power (kW)	P	1
Input voltage (V)	Vin	300 - 400
Output voltage (V)	Vo	48
Resonant frequency of Lr+Lm and Cr (kHz)	f1	243
Resonant frequency of Lr and Cr (MHz)	f2	1
Transformer voltage (V)	V_T	200
Primary current (rms) (A)	Ipri	6
Turns ratio (pri: sec1: sec2)	Tp: Ts1: Ts2	4:1:1
Magnetizing inductance (uH)	Lm	23.52
Resonant inductance (uH)	Lr	1.47
Resonant capacitance (nF)	Cr	17.2

IV. Design Approach For Integrated L-L-C-T

The design of the integrated L-L-C-T component consists of magnetic core design, winding design and dielectric design.

The first parameter to be determined is the core cross-sectional area, which can be calculated from (1).

$$A_{core} = \frac{V_T}{k_t B_T f_l T_p} \tag{1}$$

The only free variables in determining the core area are the total number of primary turns T_p and the chosen operating flux density of the core B_T. According to former research results on optimum design of the L-L-C-T structure [5][6], the lowest and second lowest possible number of turns often yield the smallest structure. The operating flux density is usually far

below the saturation flux density of core material. It should be chosen to get an acceptable loss at operating frequency and a smaller volume of the integrated L-L-C-T component. The loss estimation will be discussed in the next part of this paper.

The shape of the core can be determined by defining a core aspect ratio: k_a, which is the ratio of the width to the length of the core center leg. The core aspect ratio will determine the shape of the integrated L-L-C-T component.

The next step is determining the conductor width through choosing an appropriate current density J. Assume the skin depth of the conductor material at the highest operating frequency is δ. The width of the conductor of the primary winding is determined by (2).

$$w_p = \frac{I_{pri}}{J \cdot \delta} \tag{2}$$

To determine the width and height of the magnetic core, another design variable: the number of primary layers M_p is needed. Then the number of turns per layer is determined by (3).

$$N_p = \frac{T_p}{M_p} \tag{3}$$

Thus the winding window width can be calculated from the above parameters. The insulation thickness between adjacent turns within a layer should also be considered.

After getting the conductor width and number of turns per layer, the capacitor plate area is set. With the required capacitance and conductor plate area fixed, the only design freedom is the relative permittivity and thickness of dielectric material. The achievable design is limited by the availability of the different dielectric materials and their dimensions. Also another aspect that must be considered is the sensitivity of the relative permittivity of dielectric materials to the changes of electric field strength and temperature. For some high permittivity dielectric materials, for example Y5V, its relative permittivity value will change appreciably as function of electric field strength and temperature [10]. For LLC resonant converter, this means a large variation of the resonant frequency. Thus these high relative permittivity dielectric materials were not chosen for this design. The ceramic dielectric material Np0 has been chosen for this integrated L-L-C-T design. Np0 is one of the most stable ceramic dielectric materials, thus the variation of resonant frequency of the converter will be limited to acceptable values for varying voltage and working temperature.

The secondary winding conductor width is calculated in a similar way as for the primary winding.

The leakage inductance of the transformer is used as the resonant inductance Lr in this design. The inherent leakage inductance within the structure is normally less than the required value. A leakage layer needs to be designed and added. The approximate calculation of the inherent leakage inductance is accomplished by determining the total window height (without the leakage layer) and assuming a one-dimensional magnetic field distribution across the window, which is dependent on the current distribution.

The last part of the design is to calculate the required air gap in the main magnetic core to realize the required magnetizing inductance.

Table II shows the properties of the different materials selected for the design of the integrated L-L-C-T component for this 1MHz 1kW LLC resonant converter.

TABLE II. CHARACTERISTICS OF MATERIALS USED IN INTEGRATED L-L-C-T COMPONENT

	Magnetic material	Leakage material	Dielectric material
Name of material	3F3	FPC C350	Np0
Parameters	Permeability	Permeability	Permittivity
Symbol	μ_r	μ_i	ε_r
Value	2000 ± 20%	9 ± 20%	85

The design parameters of the integrated L-L-C-T component are chosen based on experience and the results of loss and volume estimation, which will be discussed in the next section of this paper. Table III summarizes the design parameters of the integrated L-L-C-T component for this 1MHz 1kW design.

TABLE III. DESIGN RESULTS OF INTEGRATED L-L-C-T

Parameters	**Symbol**	**Value**
Number of primary winding	Tp	4
Number of secondary winding1	Ts1	1
Number of secondary winding 2	Ts2	1
Number of layers of primary winding	Mp	2
Number of layers of secondary winding1	Ms1	1
Number of layers of secondary winding2	Ms2	1
Operating flux density (T)	B_T	0.08
Core aspect ratio	ka	0.08
Core cross section area (mm^2)	Acore	156.3
Winding window width (mm)	Ww	21
Copper width of primary winding (mm)	Wp	9.1
Copper width of secondary winding (mm)	Ws	19
Copper thickness of primary winding (um)	hpri	66
Copper thickness of secondary windings (um)	hsec	66
Thickness of dielectric material (um)	hdie	270
Thickness of leakage layer (mm)	hleak	1.5
Total length (mm)	Lt	87
Total width (mm)	Wt	50
Total height (mm)	Ht	5.9

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V. LOSS ESTIMATION

The total power loss of the integrated L-L-C-T consists of three parts: magnetic core loss, dielectric loss and winding conduction loss. Loss estimation result can be used to facilitate the thermal design of the integrated L-L-C-T component.

A. Magnetic core loss

Since the voltage waveform applied to the L-L-C-T module is not sinusoidal, the Modified Steinmetz Equation (MSE) has to be used to calculate the magnetic core loss [11]. By determining the average dB/dt over one cycle, and normalizing it with respect to a sinusoidal waveform, an equivalent frequency can be calculated and used to determine the magnetization loss per cycle. The equation to calculate the core loss density and equivalent frequency are given in (4) and (5).

$$P_v = C_m f_{eq}^{\alpha-1} \hat{B}^{\beta} f \tag{4}$$

$$f_{eq} = \frac{2}{\Delta B^2 \pi^2} \int_0^T \left(\frac{dB}{dt} \right)^2 dt \tag{5}$$

B. Dielectric loss

Similar to magnetic material, dielectric material also has a D-E hysteresis loop and its loss is a function of applied electric field strength, frequency and temperature. The dielectric loss per volume can be estimated using (6) [12].

$$P_{vol} = E^2 \omega \varepsilon_0 \varepsilon_r' \tan \delta \tag{6}$$

C. Conduction loss

Conduction loss is a major part of the total losses of an integrated L-L-C-T component. The L-L-C-T structure is quite different from a normal transformer structure in terms of current and magnetic field distribution, since the integrated capacitance adds an additional displacement current component, which is orthogonal to the conductive current as shown in Fig. 5. To get the conduction loss, both skin effect loss and proximity effect loss need to be taken into account, and these two effects can be assumed to be mathematically orthogonal to each other under certain conditions [13].

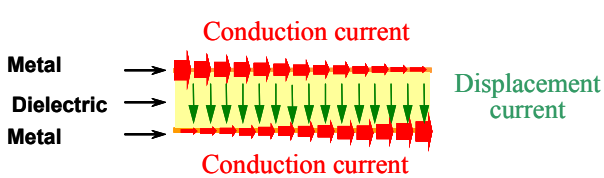


Fig. 5 Current distribution in the primary winding

After getting the current distribution within the conductors and the magnetic field distribution within the winding window, the skin effect loss and proximity effect loss can be calculated using the following equations (7) and (8).

$$P_{Skin} = \int_0^l \frac{I^2(x)}{4w\sigma\delta} \cdot \frac{\sinh\frac{h}{\delta} + \sin\frac{h}{\delta}}{\cosh\frac{h}{\delta} - \cos\frac{h}{\delta}} dx \tag{7}$$

$$P_{proximity} = \int_0^l \frac{H^2(x)w}{\sigma\delta} \cdot \frac{\sinh\frac{h}{\delta} - \sin\frac{h}{\delta}}{\cosh\frac{h}{\delta} + \cos\frac{h}{\delta}} dx \tag{8}$$

where h is the height of the conductor layer; δ is the skin depth of the conductor material at the operating frequency; and σ is the conductivity of the conductor material.

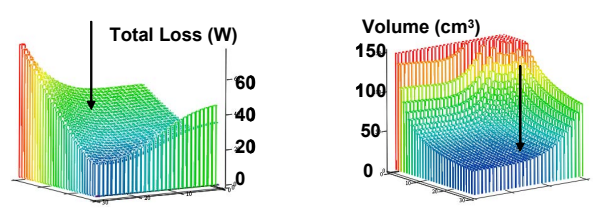
Because of the orthogonality between these two effects, they can be added up to get the total conduction loss.

After getting the total loss of the integrated L-L-C-T component, the result can be used. There are mainly 3 design variables: the magnetic core flux density B_T, the current density J and the core aspect ratio ka. To see the relationship of the total loss or volume of the integrated L-L-C-T component to these three input variables, a 4-D graph is needed. Thus we can keep one input as a constant and see the relationship of total loss and volume to the other 2 input variables. The graphs are shown in Fig. 6-8.

For Fig. 6, current density is constant: J=10A/mm^2. The relationship of total loss and volume of the integrated L-L-C-T to the other 2 input variables magnetic flux density B_T and core aspect ratio ka is shown. The arrow points out the present design.

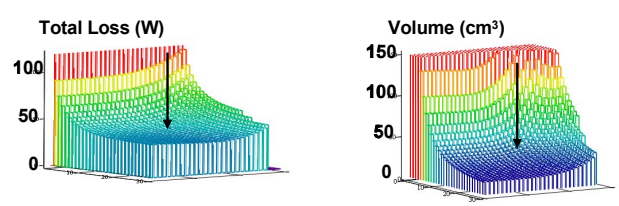
For Fig. 7, magnetic flux density is constant B_T=80mT. For Fig. 8, core aspect ratio ka=0.08.

From consideration of these graphs, a better design with less total loss and smaller volume can be chosen.



a. Total loss vs. B_T, ka b. Volume vs. B_T, ka

Fig. 6 Total loss and volume vs. B_T, ka (J=10A/mm^2)



a. Total loss vs. J, ka b. Volume vs. J, ka

Fig. 7 Total loss and volume vs. J, ka (B=80mT)

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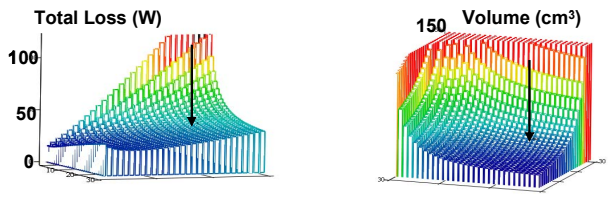
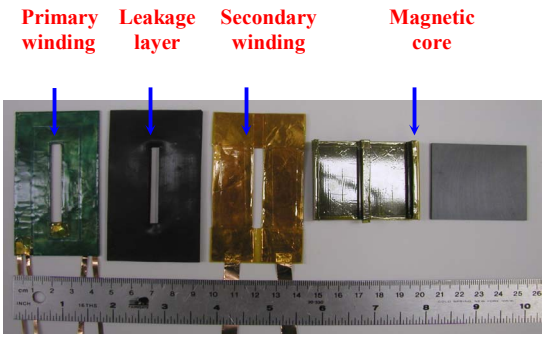


a. Total loss vs. B_T, J b. Volume vs. B_T, J

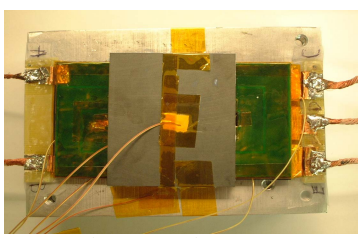
Fig. 8 Total loss and volume vs. B_T, J (ka=0.08)

VI. Prototype and Test Results

The prototype of the integrated L-L-C-T component has been constructed, as shown in Fig. 9. The figure also indicates the instrumentation with thermal couples to validate the thermal design.



a. Exploded view



b. Integrated component

Fig. 9 Prototype of integrated L-L-C-T component

The component has been tested under small signal conditions using an impedance analyzer. As shown in Fig. 10, A and B are measurement terminals. When the secondary side is open circuited, it is equivalent to Lr+Lm in series with Cr. The impedance and phase graph is shown in Fig. 11a. When C and E is short circuited, it is equivalent to Lr in series with Cr. The impedance and phase graphs are shown in Fig. 11b.

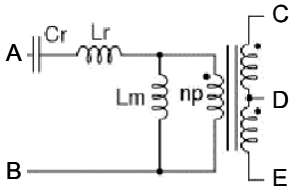
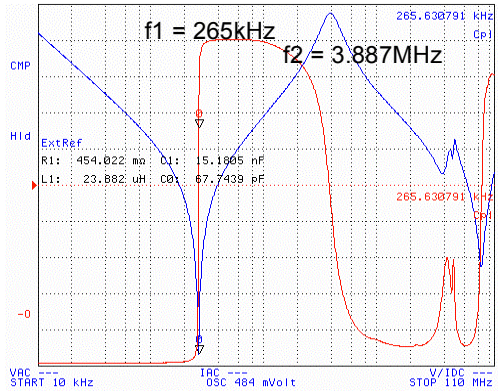
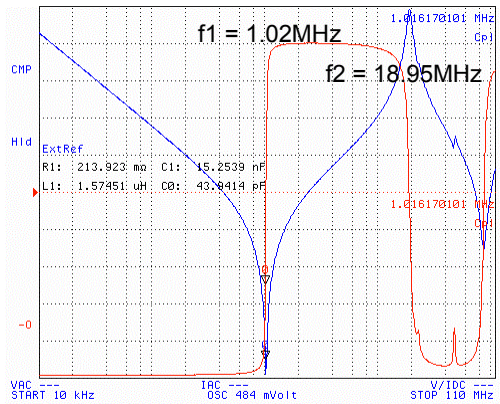


Fig. 10 Schematic of integrated L-L-C-T component



a. Impedance and phase graph with CE open circuit



b. Impedance and phase graph with CE short circuit

Fig. 11 Small signal test result using impedance analyzer

Equivalent inductance and capacitance values can be calculated from the small signal test result and they are compared with the design values in the following table. We can see that the measurement values agree well with the design values.

TABLE IV. Comparison of Measurement Value and Design Value

	Design value	Measurement value
Resonant Inductance Lr (uH)	1.47	1.57
Resonant Capacitance Cr (nF)	17.2	15.25
Magnetizing inductance Lm (uH)	23.52	22.31
Resonant frequency f1 (kHz)	243	256
Resonant frequency f2 (MHz)	1	1.02

An LLC resonant converter circuit was also constructed and the integrated L-L-C-T component was tested in-circuit.

Fig. 13 shows the waveforms of measured input voltage V_{AB}, input current I_{pri}, output voltage V_{CD} and output current I_{sec} of the resonant tank. This measurement is done under half input voltage and half load condition. The power level of the test is around 200W. Fig. 13a shows the waveforms for discrete passive components and Fig. 13b shows the waveforms for integrated L-L-C-T component. The measured efficiency of the integrated L-L-C-T component under this working condition is approximately 90.5% at a frequency of 690kHz.

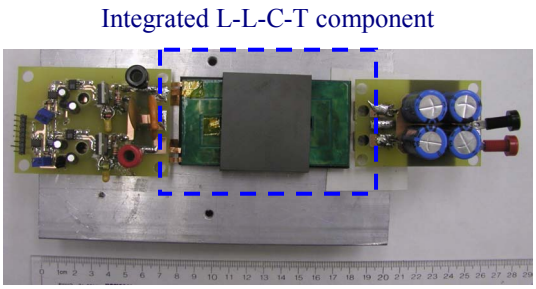
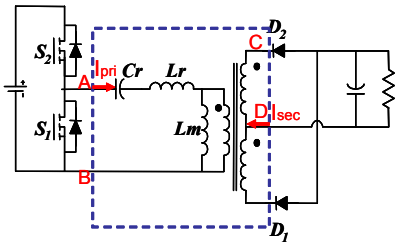
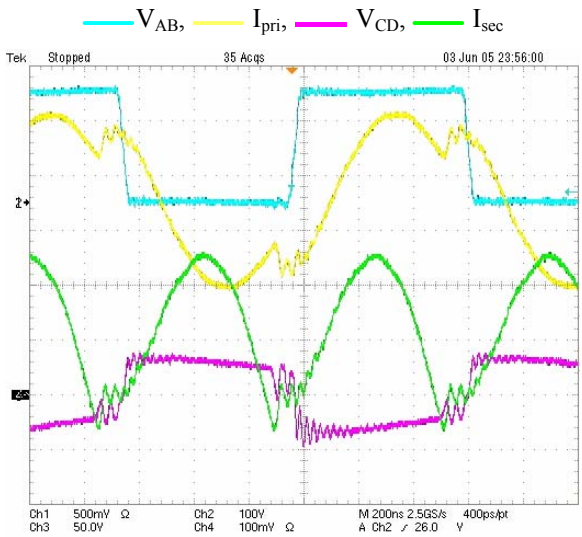
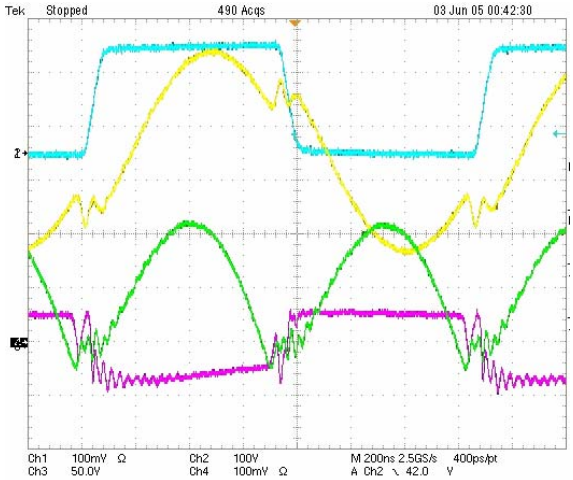


Fig. 12 Integrated L-L-C-T component in LLC resonant converter circuit





a. Waveforms with discrete passive components



b. Waveforms with integrated L-L-C-T component

Fig. 13 Waveforms of In-circuit test

VII. Conclusion

The LLC resonant converter is an excellent candidate for front-end DC/DC conversion for distributed power systems. To achieve higher power density, better performance and higher reliability, passive integration has been used to integrate all passive components within the resonant tank of the converter into one single component – an integrated L-L-C-T. The design approach and loss estimation of the integrated component have been discussed. A prototype has been constructed and small signal test results show that the measurement values agree well with the design values. In-circuit test waveforms of the integrated L-L-C-T component are also given and compared with waveforms of discrete passive components. The efficiency of the integrated L-L-C-T component is approximately 90.5% at frequency 690 kHz.

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